Expectation

math-foundations / 25-expectation

1 Definitions

Definition 1 (Expectation, discrete). *Let X be a discrete random variable on a probability space $(\Omega, \mathcal{F}, P)$ taking values $\{k_i\}_{i\geq 1} \subset \mathbb{R}$ with probability mass function $p_X(k) = P(X = k)$. If $\sum_i |k_i|\, p_X(k_i) < \infty$, the* expectation *of X is*

$$E[X] \;=\; \sum_i k_i\, p_X(k_i).$$

Definition 2 (Expectation, continuous). *Let X be a continuous random variable with probability density function f_X. If $\int_{-\infty}^{\infty} |x|\, f_X(x)\, dx < \infty$, then*

$$E[X] \;=\; \int_{-\infty}^{\infty} x\, f_X(x)\, dx.$$

Definition 3 (Expectation, Lebesgue). *For an integrable measurable $X : \Omega \to \mathbb{R}$,*

$$E[X] \;=\; \int_{\Omega} X(\omega)\, dP(\omega).$$

This is constructed first for simple functions, extended to non-negative measurable functions via monotone approximation, and to signed functions via $X = X^+ - X^-$.

2 Linearity of Expectation

Theorem 1 (Linearity of Expectation). *Let X and Y be random variables on a common probability space with $E[|X|] < \infty$ and $E[|Y|] < \infty$, and let $a, b \in \mathbb{R}$. Then*

$$E[aX + bY] \;=\; a\, E[X] + b\, E[Y].$$

Linearity holds whether or not X and Y are independent.

Proof (discrete case). Assume X and Y are discrete random variables with joint pmf $p_{X,Y}(x, y) = P(X = x, Y = y)$, and that $\sum_{x,y} |x|\, p_{X,Y}(x, y) < \infty$ and likewise for Y. Let $Z = aX + bY$.

By LOTUS applied to the joint pmf,

$$\begin{aligned} E[Z] \;=\; E[aX + bY] &= \sum_x \sum_y (ax + by)\, p_{X,Y}(x, y) \\ &= a \sum_x \sum_y x\, p_{X,Y}(x, y) \;+\; b \sum_x \sum_y y\, p_{X,Y}(x, y). \end{aligned}$$

Absolute convergence (from the integrability assumptions) permits us to rearrange the double sum and apply Fubini for series. Marginalising the joint pmf gives $\sum_y p_{X,Y}(x,y) = p_X(x)$ and $\sum_x p_{X,Y}(x,y) = p_Y(y)$, so

$$\begin{aligned} E[Z] &= a\sum_x x\left(\sum_y p_{X,Y}(x,y)\right) + b\sum_y y\left(\sum_x p_{X,Y}(x,y)\right) \\ &= a\sum_x x\, p_X(x) + b\sum_y y\, p_Y(y) \\ &= a\,E[X] + b\,E[Y]. \end{aligned}$$

No use of independence was required: the joint pmf $p_{X,Y}$ carries any dependence structure, and it factors out of each single-variable sum only through its marginals. □

Proposition 1 (Jensen's inequality). *Let $\phi : \mathbb{R} \to \mathbb{R}$ be convex and let X be an integrable random variable with $E[|\phi(X)|] < \infty$. Then*

$$\phi\big(E[X]\big) \;\le\; E\big[\phi(X)\big].$$

Proposition 2 (LOTUS). *For measurable $g : \mathbb{R} \to \mathbb{R}$ such that $g(X)$ is integrable,*

$$E[g(X)] \;=\; \int_{\mathbb{R}} g(x)\, f_X(x)\, dx \quad \textit{(continuous)}, \qquad E[g(X)] \;=\; \sum_k g(k)\, p_X(k) \quad \textit{(discrete)}.$$

3 Examples

Bernoulli. If $X \sim \text{Bernoulli}(p)$, then $E[X] = 1 \cdot p + 0 \cdot (1-p) = p$.

Uniform $[a,b]$**.** If $X \sim \text{Uniform}[a,b]$, then

$$E[X] = \int_a^b \frac{x}{b-a}\, dx = \frac{a+b}{2}.$$

Binomial. If $X \sim \text{Binomial}(n,p)$, write $X = \sum_{i=1}^n X_i$ with $X_i \sim \text{Bernoulli}(p)$ iid. By Theorem **??**, $E[X] = \sum_{i=1}^n E[X_i] = np$.